

SILENT FOLDS: PROJECTIVE POLES AS A TRAINING-FREE FAILURE FLAG IN HOMOGRAPHY-BASED RETINAL REGISTRATION

Felix Bergmann

University of Amsterdam, Amsterdam, The Netherlands
felix.bergmann2@student.uva.nl

ABSTRACT

Homographies remain the default transform in feature-based retinal registration, yet every homography has a pole line on which its denominator vanishes. Where that line crosses the fundus the warp folds, while the matrix still looks valid. Auditing 625 FIRE and COph100 pairs registered with XFeat, SuperPoint/LightGlue, and the released retina-specific SuperRetina, we find that all 100 image crossings are failures, 25 of them not flagged by the estimator, and 167 of the 168 estimates within one image diagonal of a pole fail; every crossing still fails at twice the error threshold. The pole clearance ρ , a unit-free distance from a few denominator evaluations, is a training-free quality-control (QC) score (area under the ROC curve, AUROC, 0.78–0.85 at the study endpoint). Fused with RANSAC inlier ratio it reaches 0.85–0.86, significantly above inlier ratio for two of three pipelines and not significantly different from a learned detector, and it stays at or above 0.75 across endpoints from 0.0025 to 0.02 of the image diagonal, where either score alone drops to 0.47–0.56 on some pipeline. The rectangle crossing test is equivalent to the common projected-corner check; we extend it to circular and convex supports. Poles also corrupt learned QC: two pole estimates were involved in 19 of 20 Platt-calibration reversals of a detector across 21 regroupings. A guard on the feature’s own support removes all of them and lowers the worst-case Brier score; a guard on a narrower support removes none.

Index Terms— retinal image registration, quality control, homography, failure detection, calibration

1. INTRODUCTION

Registration of fundus photographs underlies longitudinal monitoring and screening, for example of retinopathy of prematurity in infants [1, 2]. A silently failed registration corrupts every downstream change measurement, so at scale failures must be caught automatically. Feature-based retinal methods still estimate a homography: SuperRetina fits one to learned keypoints [3], and SuperJunction uses one as its first stage [4]. Registration quality control (QC), in turn, predicts failure from appearance, match, and transformation features such as Jacobians and bending energy [5, 6, 7].

A homography is a rational map. Where its projective denominator changes sign it sends image points through infinity and reverses orientation, a *fold*: in a fundus overlay, the moving image is torn along the pole line and the part beyond it is mirrored onto the far side. Close to that line its derivatives are unbounded. Robust estimators do not test for this: RANSAC scores reprojection consensus, and the returned matrix is finite. Moisan et al. reject such “wild homographies” by enforcing orientation preservation at the four sample points [8], which constrains those points but not the rest of the image.

Contributions. (1) An audit of three modern retinal pipelines on FIRE and COph100 [9, 1], including the released SuperRetina: pole crossings are always failures, often silent ones, and pole distance is a training-free QC score that adds to inlier ratio. (2) Tools: the rectangle crossing test is algebraically the common projected-corner convexity check; we extend it to circular and convex supports, normalize the pole distance into a scale-invariant clearance, and apply it as a RANSAC hypothesis constraint. (3) A cautionary finding for learned QC: homography-derived derivative features are unbounded, so a detector must guard, transform, or drop them, and a guard must contain the feature’s support.

2. METHOD

Pole and folding. A homography H maps (x, y) to

$$u = \frac{h_{11}x + h_{12}y + h_{13}}{d}, \quad v = \frac{h_{21}x + h_{22}y + h_{23}}{d}, \quad (1)$$

with denominator $d(x, y) = h_{31}x + h_{32}y + h_{33}$ and Jacobian determinant $\det J(x, y) = \det H / d(x, y)^3$ [10]. Hence $\det J$ changes sign exactly where d does: the pole line $d = 0$ is the fold, and $|\det J|$, like every higher derivative, diverges as $d \rightarrow 0$.

Exact support tests. Because d is affine, its extrema on an axis-aligned rectangle lie at the four corners c_k , so the pole crosses the closed rectangle if and only if

$$\min_k d(c_k) \leq 0 \leq \max_k d(c_k). \quad (2)$$

This is the algebraic form of the common check that the projected corners form a convex quadrilateral with the source orientation: a corner triangle’s image orientation is $\text{sign}(\det H) \text{sign}(d_i d_j d_k)$, so the two agree unless a corner lies on the pole. Unlike the projected check, Eq. (2) extends to any convex support, such as a fundus mask’s convex hull, by evaluating d at its vertices (for a non-convex mask this is conservative). For a circular field of view (FOV) with center q and radius r , with $a = (h_{31}, h_{32})$, the condition is $|a^\top q + h_{33}| \leq r \|a\|_2$; we use the image’s inscribed circle as the FOV proxy. H maps moving to fixed image and is tested on the moving support; H^{-1} is tested on the fixed image.

Clearance. For a non-crossing rectangle with diagonal L ,

$$\rho(H) = \min_k |d(c_k)| / (L \|a\|_2) \quad (3)$$

is the distance from the image to the pole line in image diagonals; $\rho = 0$ for a crossing and $\rho = \infty$ for an affine map. It is invariant to matrix scale and image resolution and costs four denominator evaluations.

Uses. (i) *QC score*: $s = -\log_{10} \max(\rho, 10^{-3})$, with ρ capped at 10^3 ; the *fused* score averages the rank percentiles of s and of

Case 1: pole clips a corner, misses the FOV

Case 2: pole crosses the FOV

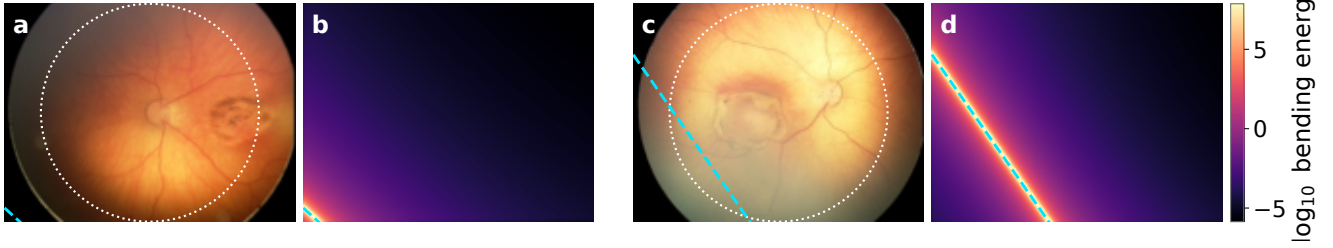


Fig. 1. Two failed Coph100 XFeat-H registrations (patient 055). (a,c) Moving fundus images; (b,d) log bending-energy density of the estimated warp on one shared scale. Cyan: pole line $d = 0$; dotted: inscribed-FOV proxy. In case 1 the pole clips the rectangle but misses the FOV and the fundus mask; in case 2 it crosses the FOV along a 294.9-pixel chord. Images: RIDIRP (CC0); annotations: Coph100 (CC BY 4.0) [2, 1].

Table 1. Pole audit on 625 retinal pairs. Failed / flagged: returned transforms that fail among those each criterion flags. AUROCs are pooled and group-weighted over all attempted pairs, with 95% group-bootstrap intervals; inlier ratio is a pre-specified training-free baseline (for SuperRetina, its LMedS inlier rate); fused: mean rank percentile of $-\log \rho$ and inlier ratio; detector: median raw AUROC of the guarded Base detector over 21 assignments. Constrained RANSAC (rectangle): failures made successes / silent failures made explicit / explicit failures made silent; for SuperRetina, the study RANSAC re-fits its stage-1 matches. SIFT-H is a weak-matching stress test.

Pipeline	Returned	Failed	Failed / flagged			AUROC				Constr.
			Rect. cross.	FOV cross.	$\rho \leq 1$	$-\log \rho$	Inlier ratio	Fused	Detector	RANSAC
XFeat-H	625	342	2/2	1/1	20/21	.778 (.71-.84)	.848 (.80-.90)	.855 (.80-.90)	.833	1/0/0
SPLG-H	618	278	10/10	6/6	40/40	.823 (.77-.87)	.782 (.71-.85)	.846 (.79-.90)	.838	0/4/0
SuperRetina	568	277	88/88	81/81	107/107	.852 (.80-.90)	.811 (.75-.87)	.856 (.80-.91)	–	29/2/72
SIFT-H	521	519	375/375	187/187	412/412	–	–	–	–	0/0/104

the negated RANSAC inlier ratio. Neither is trained. (ii) *Feature guard*: a derivative feature is set missing when the pole crosses the support on which it is computed. (iii) *Support-constrained RANSAC*: minimal-sample hypotheses whose pole crosses the support are discarded before consensus scoring, on the same random stream; if the consensus refit crosses, the best admissible hypothesis is returned, and if none exists the registration fails explicitly.

3. EXPERIMENTAL SETUP

Data and endpoint. We use 134 FIRE [9] and 491 Coph100 [1] pairs, grouped into 42 image-connected FIRE components (patient identifiers are unavailable) and 56 Coph100 patients. A registration fails if it returns no transform (an *explicit* failure) or its mean landmark error exceeds 0.005 of the fixed-image diagonal in original coordinates (a *silent* failure): 20.6 pixels on FIRE (2912²), within its standard 0–25-pixel success curve, and 4 pixels on Coph100 (640 × 480).

Pipelines. The generic (not retina-specific) pipelines XFeat-H, SPLG-H, and SIFT-H match XFeat [11], SuperPoint/LightGlue [12], or SIFT [13] correspondences at a 1,024-pixel long edge (Coph100 upsampled 1.6×) and share one seeded RANSAC with a normalized DLT (3-px threshold, 10⁴ iterations, consensus refit). SuperRetina is run as released (least-median-of-squares, LMedS, with its two-pass refinement) at original resolution [3]; it reproduces the published FIRE success-curve AUCs of the S/P/A (large-overlap/small-overlap/anatomical-change) categories within 0.011.

Learned detector. Per pipeline, an L2-regularized logistic regression combines image, match, inverse-consistency, geometric (including bending energy $\mathcal{B}$ on a 32 × 32 lattice, our curvature feature), cross-pipeline, and structure features (Base); a second arm adds re-

sampling and perturbation stability (+Stab). Features are standardized after training-median imputation with one missingness indicator each, and $C \in \{0.1, 1, 10\}$ is chosen by grouped inner cross-validation. Platt scaling [14], an unregularized logistic fit of the raw score, uses a disjoint calibration fold; a negative slope reverses the ranking. With five group folds (test f , calibration $f+1$, training the rest), we refit everything under 21 seeded group-to-fold assignments and pool test predictions with equal total weight per patient group. Assignment spread reuses cases and is not a confidence interval; intervals are 95% group bootstraps (2,000 resamples of patients or components) on the original assignment. No-output pairs receive the most-failing value of every training-free score.

4. RESULTS

4.1. Pole crossings are failures, often silent

All 12 crossings by the generic matchers are failures (Table 1); they come from four Coph100 patients and none from FIRE. SuperRetina with its own LMedS returns 88 crossing homographies from 38 Coph100 patients, all failures; 75 have no LMedS inliers, which its evaluation counts as failed, but 13 would pass silently. Testing H^{-1} on the fixed image adds 4 SPLG-H crossings (4 failed); none of the estimates is a global reflection. The projected-corner check flags the same crossings, as it must; the 32 × 32 lattice det J test costs 256 times more and misses 3/375 SIFT-H crossings. Under weak matching, invalid geometry is common: 72% of SIFT-H homographies cross the image. Crossings alone have low recall (12 of 620 generic-matcher failures); the clearance extends them: within one diagonal, 60 of 61 generic-matcher and all 107 SuperRetina estimates fail (Fig. 2b). This does not depend on the endpoint: all crossings

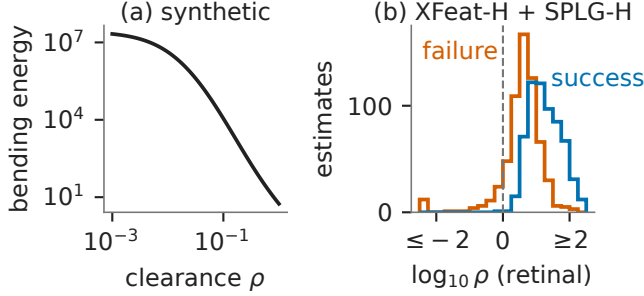


Fig. 2. (a) Bending energy on the 32×32 lattice for a synthetic homography whose pole approaches a 640×480 image from outside. (b) Clearance of all returned XFeat-H and SPLG-H estimates; ≤ -2 holds the 12 crossings. Dashed: $\rho = 1$, left of which 60/61 estimates fail.

fail up to 0.01 of the diagonal and 99/100 at 0.02, and restricting to returned transforms changes no AUROC below by more than 0.03.

4.2. Clearance is a training-free QC score

$-\log \rho$ reaches AUROC 0.778/0.823/0.852 for XFeat-H/SPLG-H/SuperRetina (Table 1). Inlier ratio is stronger on XFeat-H (0.848) and weaker on the others (0.782/0.811). The two are complementary: the fused score exceeds inlier ratio on SPLG-H and SuperRetina (paired 95% intervals of the AUROC difference [0.03, 0.11] and [0.01, 0.08]) but not on XFeat-H ($[-0.03, 0.04]$), and is not significantly different from the learned detector ($[-0.01, 0.07]$, $[-0.04, 0.06]$). On SuperRetina estimates with at least one inlier (482), ρ still reaches 0.725 versus 0.669. The pole distance from the image center ranks identically (differences within ± 0.004): the exact support matters for the crossing decision and the unit threshold, not for ranking. A warped-area scale baseline reaches only 0.68/0.70 (XFeat-H/SPLG-H). Flagging $\rho \leq \tau$ with τ chosen on training groups for $\geq 95\%$ precision keeps 92%/94% held-out precision at 20%/35% recall (median $\tau = 2.34/2.93$ diagonals). At looser endpoints (0.01/0.02 of the diagonal), the AUROC of ρ rises for SPLG-H and SuperRetina (0.90/0.94, 0.95/0.99) but falls for XFeat-H (0.70/0.47), whose near-pole estimates are often only moderately wrong, while inlier ratio degrades on SPLG-H. The fused score is the robust choice: across endpoints from 0.0025 to 0.02 its lowest AUROC is 0.75, versus 0.47 for ρ and 0.56 for inlier ratio.

4.3. Poles corrupt learned calibration

In one held-out fold (XFeat-H, +Stab, original assignment), the raw detector ranks failures well (AUROC 0.927), but its Platt slope is -4.3×10^{-5} , compressing all 83 test probabilities into [0.57961, 0.57970] and reversing them (0.073). The calibration fold held the two patient-055 poles of Fig. 1, whose bending energies (8.65×10^9 , 1.05×10^9) lie orders of magnitude outside the training range; deleting both restores the fit, deleting either alone does not. The 20 reversals across 21 assignments repeat this mechanism rather than adding independent evidence: 19 reuse these two estimates, and one SPLG-H fold holds six poles from two patients. The cause is geometric: moving a synthetic pole from $\rho = 1$ to 10^{-3} outside a 640×480 image raises lattice bending energy from 5.53 to 2.13×10^7 (Fig. 2a).

Homography-derived derivative features are therefore unbounded, and a detector must guard, transform, or drop them. All three prevent reversal and lower the worst-assignment Brier score (Table 2); log

Table 2. Calibration over 21 assignments. Reversals (105 folds per cell) for Base/+Stab; XFeat-H Base calibrated AUROC, Brier score, and NLL as median (worst assignment). The curvature and guard supports are crossed; “none” is unguarded. SPLG-H worst Brier: .175 unguarded, .167 guarded. FOV/FOV and removal differ per case but agree to three decimals. [†]One fit with an optimizer warning.

Curv.	Guard	Reversals		XFeat-H Base		
		supp.	supp.	XFeat-H	SPLG-H	AUROC Brier NLL
Rect.	none	13 [†] /6	1/0	.787	.180 (.206)	.547 (.603)
Rect.	FOV	13 [†] /6	1/0	.788	.180 (.207)	.539 (.599)
Rect.	Rect.	0/0	0/0	.828	.169 (.179)	.518 (.559)
FOV	none	11/6	0/0	.802	.178 (.207)	.536 (.599)
FOV	FOV	0/0	0/0	.828	.169 (.181)	.514 (.559)
Isotonic		0/0	0/0	.818	.171 (.184)	.968 (1.57)
Quantile curv.		0/0	0/0	.835	.165 (.177)	.509 (.554)
Remove curv.		0/0	0/0	.828	.169 (.181)	.514 (.559)

transforms and clipping behave like removal, and isotonic calibration [15] prevents reversal but inflates NLL. A guard must contain the feature’s support: sampling curvature only inside the FOV leaves 17 reversals, because the pole can cross the FOV itself, and an FOV guard on rectangle-sampled curvature removes none, because a pole can clip the rectangle while missing the FOV (Fig. 1a); over-guarding is harmless. Beyond preventing reversal, the fixes are practically equivalent here: support-matched guards and removal give different per-case predictions but agree within 0.004 in every metric, so once the poles are neutralized curvature adds little signal, and the detector does not profit from the guard’s missingness flag. The guard’s value is diagnostic: it keeps curvature where it is valid and records the pole as an explicit flag, one that marked a failure in 100 of 100 crossings, whereas removal and monotone transforms hide it.

4.4. Support-constrained RANSAC

Moisan’s sample-point test [8] still returns 13 crossings (2 XFeat-H, 11 SPLG-H), since a minimal sample can lie on one side of a pole that crosses the image; the support constraint returns none, but fold-free is not correct. Of the 12 generic-matcher pole cases it makes 4 explicit failures and 8 admissible wrong estimates, and under weak matching it replaces explicit failures with wrong estimates (72 SuperRetina, 104 SIFT-H pairs; Table 1), so its output still needs QC.

5. DISCUSSION AND CONCLUSION

A finite homography is not a valid warp of the fundus. Released retinal pipelines return pole-crossing homographies, some silently, and every one we observed was a failure. A few denominator evaluations detect them exactly on rectangular, circular, or convex supports, and the clearance turns the same quantity into a training-free QC score that adds to inlier ratio; fused, it reaches the observed accuracy of a learned detector without training. Learned QC must guard, transform, or drop homography-derived derivative features; these fixes perform alike, and a support-containing guard additionally keeps the pole as an explicit flag. Constraining RANSAC removes folds but not errors. In practice we recommend three steps for any homography-based retinal pipeline: reject estimates whose pole crosses the fundus support; report the clearance ρ next to the inlier ratio, or fuse the two, as a QC score; and guard derivative features on the support on which they are computed. The study is limited to two public retinal datasets,

generic-matcher poles from four patients (SuperRetina’s span 38), FIRE groups that may share unobserved patients, and homographies. Quadratic retinal models [16, 17] and eye-model registration [18] have no projective denominator; homographies persist in learned pipelines because they are estimable from four points, and they are applied even to infant retinopathy-of-prematurity imaging, where all poles of the three modern pipelines in our data occur.

6. COMPLIANCE WITH ETHICAL STANDARDS

This research study was conducted retrospectively using human subject data made available in open access by FIRE [9], RIDIRP [2], and COph100 [1]. FIRE reports written informed consent; RIDIRP reports consent and approval by the Ethical Committee of University Hospital Ostrava (544/2018). This secondary analysis used only the publicly released, de-identified data; no new participant data were collected.

7. ACKNOWLEDGMENTS

No external funding; the author declares no conflicts of interest. Anthropic Claude and OpenAI Codex generated parts of the analysis code (Sections 3–4) and assisted with drafting and editing Sections 1–5; the author verified all code, numbers, citations, and text. Code and per-case results: <https://github.com/Felix-Bergmann-Code/silent-folds>.

8. REFERENCES

- [1] Yan Hu, Mingdao Gong, Zhongxi Qiu, Jiabao Liu, Hongli Shen, Mingzhen Yuan, Xiaoqing Zhang, Heng Li, Hai Lu, and Jiang Liu, “COph100: A comprehensive fundus image registration dataset from infants constituting the RIDIRP database,” *Scientific Data*, vol. 12, pp. 99, 2025.
- [2] Juraj Timkovič, Jana Nowaková, Jan Kubíček, Martin Hasal, Alice Varyšová, Lukáš Kolarčík, Kristýna Maršolková, Martin Augustynek, and Václav Snášel, “Retinal image dataset of infants and retinopathy of prematurity (RIDIRP),” *Scientific Data*, vol. 11, pp. 814, 2024.
- [3] Jiazhen Liu, Xirong Li, Qijie Wei, Jie Xu, and Dayong Ding, “Semi-supervised keypoint detector and descriptor for retinal image matching,” in *European Conference on Computer Vision*, 2022, pp. 593–609.
- [4] Yu Wang, Xiaoye Wang, Zaiwang Gu, Weide Liu, Wee Siong Ng, Weimin Huang, and Jun Cheng, “SuperJunction: Learning-based junction detection for retinal image registration,” *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, no. 1, pp. 292–300, 2024.
- [5] Joshua Bierbrier, Houssein-Eddine Gueziri, and D. Louis Collins, “Estimating medical image registration error and confidence: A taxonomy and scoping review,” *Medical Image Analysis*, vol. 81, pp. 102531, 2022.
- [6] Hessam Sokooti, Gorkem Saygili, Ben Glocker, Boudewijn P. F. Lelieveldt, and Marius Staring, “Quantitative error prediction of medical image registration using regression forests,” *Medical Image Analysis*, vol. 56, pp. 110–121, 2019.
- [7] M. Elena Martinez-Perez, Franziska G. Rauscher, Pingping Zhao, and Tobias Elze, “Development and evaluation of customized software to automatically align macula and optic disc centered scanning laser ophthalmoscope fundus images,” *PeerJ Computer Science*, vol. 11, pp. e2621, 2025.
- [8] Lionel Moisan, Pierre Moulon, and Pascal Monasse, “Automatic homographic registration of a pair of images, with a contrario elimination of outliers,” *Image Processing On Line*, vol. 2, pp. 56–73, 2012.
- [9] Carlos Hernandez-Matas, Xenophon Zabulis, Areti Triantafyllou, Panagiota Anyfanti, Stella Douma, and Antonis A. Argyros, “FIRE: Fundus image registration dataset,” *Journal for Modeling in Ophthalmology*, vol. 1, no. 4, pp. 16–28, 2017.
- [10] Richard Hartley and Andrew Zisserman, *Multiple View Geometry in Computer Vision*, Cambridge University Press, 2nd edition, 2004.
- [11] Guilherme Potje, Felipe Cadar, André Araujo, Renato Martins, and Erickson R. Nascimento, “XFeat: Accelerated features for lightweight image matching,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 2682–2691.
- [12] Philipp Lindenberger, Paul-Edouard Sarlin, and Marc Pollefeys, “LightGlue: Local feature matching at light speed,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 17627–17638.
- [13] David G. Lowe, “Distinctive image features from scale-invariant keypoints,” *International Journal of Computer Vision*, vol. 60, no. 2, pp. 91–110, 2004.
- [14] John C. Platt, “Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods,” in *Advances in Large Margin Classifiers*, pp. 61–74. MIT Press, 1999.
- [15] Bianca Zadrozny and Charles Elkan, “Transforming classifier scores into accurate multiclass probability estimates,” in *Proceedings of the Eighth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2002, pp. 694–699.
- [16] Ali Can, Charles V. Stewart, Badrinath Roysam, and Howard L. Tanenbaum, “A feature-based, robust, hierarchical algorithm for registering pairs of images of the curved human retina,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 24, no. 3, pp. 347–364, 2002.
- [17] Charles V. Stewart, Chia-Ling Tsai, and Badrinath Roysam, “The dual-bootstrap iterative closest point algorithm with application to retinal image registration,” *IEEE Transactions on Medical Imaging*, vol. 22, no. 11, pp. 1379–1394, 2003.
- [18] Carlos Hernandez-Matas, Xenophon Zabulis, and Antonis A. Argyros, “REMPE: Registration of retinal images through eye modelling and pose estimation,” *IEEE Journal of Biomedical and Health Informatics*, vol. 24, no. 12, pp. 3362–3373, 2020.